

Discussion 5 Worksheet

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Problem 1: For each of the following sets V , determine whether V is a vector space with the operations of usual vector addition and scalar multiplication over $\mathbb{R}$. Justify your answer in each case.

(a) $V = \{(x, y) \in \mathbb{Z}^2\}$

(b) $V = \{(x, y, z, w) \in \mathbb{R}^4 \mid x + 2y - z + w = 5\}$

Problem 2: In each of the following, V is a vector space and $S \subseteq V$. Determine whether S is a valid subspace of V under the same operations as V . Justify your answer in each case.

(a) $V = \mathbb{R}^2$ and S is the x-axis.

(b) $V = \mathbb{R}^3$ and $S = \{(x, y, z) \in \mathbb{R}^3 \mid x - 2y + z = 1\}$.

(c) $V = \mathbb{R}^2$ with vector addition and scalar multiplication defined by

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2, y_1 + y_2 - 5), \quad \alpha \odot (x, y) = (\alpha x, \alpha y + (1 - \alpha)5),$$

and $S = \{(0, y) \mid y \in \mathbb{R}\}$.

Problem 3: A superhero is at the origin of a two-dimensional plane. The superhero has two devices: a hoverboard that moves any distance in the direction $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and a magic carpet that moves any distance in the direction $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

(a) An evil villain is hiding out on the plane at the point $\begin{pmatrix} -5 \\ 7 \end{pmatrix}$. How many hoverboard units and magic carpet units does the superhero have to move to get to the villain?

(b) Is there anywhere in the plane that the villain could safely hide and not be reached?

(c) The superhero and the villain are transported to a three-dimensional space where the superhero now has three devices:

$$\text{hoverboard: } \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \quad \text{magic carpet: } \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \quad \text{scooter: } \begin{pmatrix} -2 \\ 0 \\ 6 \end{pmatrix}$$

Is there anywhere that the villain could safely hide?

Problem 4: Identify whether each subset is linearly dependent or linearly independent and if they form a basis for $\mathbb{R}^3$.

(a) $\left\{ \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} \right\}$

(b) $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$

$$(c) \quad \left\{ \begin{pmatrix} 9 \\ 9 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 5 \\ -4 \end{pmatrix}, \begin{pmatrix} 12 \\ 12 \\ -1 \end{pmatrix} \right\}$$

Problem 5: Represent the vector $\vec{v}$ with respect to each of the two bases.

$$\vec{v} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \quad B_1 = \left\langle \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\rangle, \quad B_2 = \left\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \end{pmatrix} \right\rangle.$$